

HW 3: Theory

Wednesday, September 25, 2024 3:09 PM

1. [10 points] Let A be an $m \times d$ matrix, and let $X = AA^T$. Assume that X has d distinct, non-zero eigenvalues. Assume that $m \gg d$. In order to find the eigendecomposition of X , we will need to find the eigendecomposition of an $m \times m$ matrix. Since m is much larger than d , this is slow. Give an algorithm for finding the eigenvectors and eigenvalues of X that only requires computing the eigendecomposition of a $d \times d$ matrix. You can use simple matrix operations and assume that you have an eigendecomposition “black box” subroutine, but avoid using the SVD as a black box.

$A = m \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}^d, \quad X = AA^T = m \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}^m$

↖

this is like m points in $\mathbb{R}^d$

X has d distinct, non-zero eigenvalues
 $m > d$

eigenvalue decomposition: $A = V\Lambda V^{-1}$

let $Y = A^T A = d \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}^d$

from office hours:

$$Y\vec{v} = \lambda\vec{v} = (A^T A)\vec{v}$$

$$\Rightarrow A(A^T A)\vec{v} = A(\lambda\vec{v})$$

$$\Rightarrow X(A\vec{v}) = \lambda(A\vec{v})$$

$$\Rightarrow A\vec{v} \text{ is the eigenvector and } \lambda \text{ is the eigenvalue of matrix } X$$

Algo:

1. Compute $EVD(Y) = VDV^{-1}$, gives $V = \begin{bmatrix} | & & | \\ v_1 & \dots & v_d \\ | & & | \end{bmatrix}_{d \times d}$ and $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix}_{d \times d}$

2. Compute eigenvectors of X

$$v'_1 = Av_1, \dots, v'_d = Av_d$$

3. Eigenvalues of Y are eigenvalues of X

4. Assemble EVD of $X = AA^T$, padding 0's to make $m \times m$

$$X = V'D'V'^{-1} = \begin{bmatrix} V' & \\ Av_1 \dots Av_d & 0 \end{bmatrix}_{m \times m} \begin{bmatrix} D' & \\ \lambda_1 & \ddots & \lambda_d \\ 0 & & 0 \end{bmatrix}_{m \times m} \begin{bmatrix} (V')^{-1} \\ \end{bmatrix}_{m \times m}$$

2. In this problem we explore some relationships between SVD, PCA and linear regression.

- (a) [2 points] True or false: linear regression is primarily a technique of *supervised* learning, i.e. where we are trying to fit a function to labeled data.
- (b) [2 points] True or false: PCA is primarily a technique of *unsupervised* learning, i.e. where we are trying to find structure in unlabeled data.
- (c) [2 points] True or false: SVD is primarily an operation on a *dataset* whereas PCA is primarily an operation on a *matrix*.
- (d) [2 points] A common problem in linear regression is *multicollinearity*, where the input variables are themselves linearly dependent. For example, imagine a healthcare data set where height is measured both in inches and centimetres. This is a problem because there may now be multiple w satisfying $y = w \cdot x$. Explain how you could use a preprocessing step to solve this problem.

(a) True, we are using the y labels for training.

(b) True, we don't have labels, but are finding inherent structure.

(c) False, they're both operations on matrices, but if anything PCA is more of a dataset operation, identifying orthogonal axes of greatest variation over m observations in d attributes.

(d) Dimensionality reduction:

PCA could be used to preprocess the data, ensuring we are modeling orthogonal components, i.e. no multicollinearity.

Removing features:

Remove the redundant collinear features. We can identify these with a correlation matrix.